

FINS5557 Applied AI in Finance

Group Project — Track A: Tech Team

Assignment Type: Group Project — Track A **Weight:** 30% of final grade **Due Date:** Week 8 — submitted via the university learning management system by 11:59 PM on the Friday of Week 8 **Team Size:** 3 students

Marking Criteria: See attached **Alternative track:** [Project B — Business Team](#)

Overview

Track A teams design, build, and deploy a working AI-powered finance application. The application must incorporate at least one AI/ML technique covered in the course, be deployed to a public URL, and be demonstrated live in a 5–7 minute video.

Track selection is final once submitted to the instructor by the end of Week 4.

Learning Objectives

Upon completion of this project, students will be able to:

1. Design and implement a functional AI-powered application in a finance context
 2. Integrate machine learning models or APIs into a working software system
 3. Evaluate the ethical implications, governance obligations, and limitations of their AI system
 4. Document and present technical work to a professional standard
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Team Formation

- Teams of exactly **three students** — no exceptions without prior written approval from the course coordinator
 - Teams must be registered via UNSW Moodle by the end of **Week 3**
 - Submit a one-paragraph project pitch (problem, proposed AI technique, target user) by the end of **Week 4**; instructor sign-off is required before proceeding
 - Supervisory consultation sessions are available in Weeks 5 and 6 — book via UNSW Moodle
 - All three members are expected to contribute equitably; a **contribution statement** (one paragraph per member, signed by all three) must be included in the report appendix
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Objective

Design, develop, and deploy a functional AI-powered application that addresses a genuine problem or opportunity in finance or banking. The application must incorporate at least one AI/ML technique covered in the course and must be demonstrable — prototype quality is acceptable; theoretical wireframes are not.

Scope and Constraints

- The application must be **functional and demonstrable** at submission time
 - At least one AI/ML component must be integrated (see options below)
 - Deployment to a public URL is required — free-tier platforms are sufficient
 - The application must handle errors gracefully and include basic input validation
 - Security must be considered: no hardcoded API keys; use `.env` pattern or equivalent
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AI/ML Integration — Choose at Least One

Category	Examples
Natural language processing	Sentiment analysis, document summarisation, chatbot, named entity recognition
Predictive analytics	Credit scoring, churn prediction, fraud detection, price forecasting
Recommendation systems	Portfolio suggestions, next-best-action, product matching
Generative AI	Report generation, regulatory document drafting, financial Q&A
Computer vision	Receipt OCR, document extraction, KYC identity verification

Pre-trained models (Hugging Face, OpenAI API, Google Gemini) are permitted and encouraged. Custom-trained models on a relevant dataset attract additional marks for innovation.

Suggested Project Ideas

The following examples illustrate expected scope. Teams may propose any original idea subject to instructor approval via the Week 4 pitch.

1. **Financial Report Summariser** — NLP pipeline that ingests an ASX or SEC filing and generates a structured executive summary with sentiment scores by section
 2. **Personal Budget Advisor** — predictive model that categorises transactions, identifies anomalies, and generates natural-language savings recommendations
 3. **BNPL Credit Screener** — alternative-data credit scoring model for thin-file applicants with a Streamlit interface showing SHAP explanations for each decision
 4. **Earnings Call Sentiment Tracker** — real-time pipeline that transcribes, scores, and visualises sentiment drift across quarterly earnings calls for a chosen sector
 5. **Regulatory Document Q&A** — RAG system grounded on APRA, ASIC, or SEC regulatory PDFs, enabling plain-language queries with source citations
 6. **KYC Document Extractor** — computer vision pipeline that extracts and validates structured fields from identity documents, with confidence scores and a human-review flag
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Recommended Technology Stack

Layer	Options
Language	Python 3.10+
Framework	Streamlit, Flask, FastAPI
AI/ML libraries	scikit-learn, Hugging Face Transformers, LangChain, OpenAI SDK, Google Generative AI
Version control	GitHub (public repository, or grant instructor access before the deadline)
Deployment	Streamlit Cloud, Vercel, Hugging Face Spaces, Google Cloud Run (all free tier)

Use of AI coding assistants (Claude Code or equivalent) is permitted and must be disclosed in the written report.

Deliverables

#	Item	Format
1	GitHub repository	Public URL; source code, <code>README.md</code> , <code>requirements.txt</code>
2	Written report	PDF, ≤15 pages excluding references and appendices
3	Video demonstration	MP4 or YouTube link, 5–7 minutes

Written Report Structure

The report must follow this section sequence (15 pages maximum, body text only; references and appendices excluded from the page count):

a) Introduction — problem statement and motivation; target users and use-case scenarios; project objectives

b) Background and Literature Review — overview of relevant AI/ML techniques; review of 5–7 academic sources or technical documentation; analysis of comparable existing applications; justification for the chosen approach over alternatives

c) Methodology — system architecture diagram; AI/ML model selection and justification; dataset description (source, size, features, and any preprocessing applied); development process and tools used; challenges encountered and solutions implemented

d) Results and Evaluation — application performance metrics; AI model accuracy or effectiveness measurements; user testing results (minimum three test users, with a structured feedback instrument); limitations and areas for improvement

e) Ethical Considerations — potential biases in the AI system; privacy and data security measures; responsible AI principles applied; societal impact assessment; applicable regulatory obligations (GDPR, Privacy Act 1988, APRA CPS 234 as relevant)

f) References — minimum seven credible academic or technical sources in APA 7th or IEEE format

README.md Requirements

The README.md in the GitHub repository must include:

- Project description and full feature list
 - Installation and environment setup instructions
 - Usage guide with worked examples and screenshots
 - Dependency list (requirements.txt or equivalent)
 - API key setup instructions — describe the .env pattern without exposing actual keys
 - Known issues and planned future enhancements
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Video Demonstration (5–7 minutes)

The video must include:

- Live demonstration of the running application
- Narrated walkthrough of key features and AI integration
- Explanation of at least one technical challenge and how it was resolved
- Honest acknowledgement of one limitation or failure mode
- All three team members on camera and speaking substantively

Technical requirements:

- Minimum 1080p screen recording
 - Clear audio — use a microphone
 - Upload to YouTube (unlisted is acceptable) or submit as MP4 via UNSW Moodle
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Submission Checklist

Submit all items via UNSW Moodle by **11:59 PM, Friday of Week 8**.

- ☐ GitHub repository URL (public, or instructor access granted before the deadline)
- ☐ Written report (PDF, ≤15 pages body)
- ☐ Video demonstration (YouTube URL or MP4)
- ☐ Contribution statement (one paragraph per member, signed, in report appendix)
- ☐ AI tool disclosure statement (if Claude Code or equivalent was used, state which tasks)

Late submissions incur a penalty of 5% of the total mark per calendar day, up to a maximum of five days. Submissions more than five days late receive zero unless an extension is approved in writing by the course coordinator before the deadline.

Academic Integrity

- All code must be the original work of the submitting team
- Use of AI coding assistants must be disclosed with a description of the tasks they assisted with
- Pre-trained models and external APIs may be used but must be cited and documented
- Properly attribute all external libraries, code snippets, data sources, and figures
- Collaboration with other teams is not permitted; teams may discuss general approaches in class but must not share code or written materials
- Generative AI may be used as a drafting or ideation tool; AI-generated content must be critically reviewed and substantially revised — wholesale submission of unedited AI output constitutes academic misconduct under UNSW policy

Plagiarism or misconduct will result in a grade of zero and referral to the UNSW Academic Conduct Board.

Project Timeline

Week	Milestone
Week 3	Team of three registered via UNSW Moodle
Week 4	Track A elected; project pitch submitted; instructor sign-off required
Week 5	Optional consultation session — book via UNSW Moodle
Week 6	Optional consultation session — book via UNSW Moodle
Week 7	Draft report recommended to be complete; internal team review
Week 8	All deliverables submitted via UNSW Moodle by 11:59 PM Friday
